Chapter 1
Systems of Linear Equations



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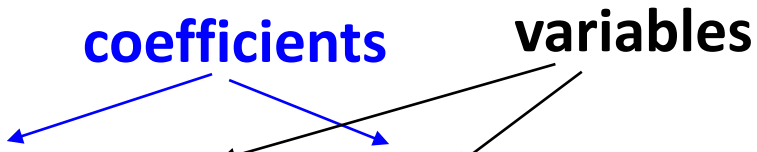
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1.1. Solutions and Elementary Operations



- $a_1x_1+a_2x_2+\ldots+a_nx_n=b$ is called a **linear equation (phương trình tuyến tính)**
- A solution (nghiệm) to the equation is a sequence $s_1,s_2,\ldots,s_n$ such that $a_1s_1+a_2s_2+\ldots+a_ns_n=b$
- $s_1,s_2,\ldots,s_n$ is called a solution to a system of linear equations if $s_1,s_2,\ldots,s_n$ is a solution to every equation of the system
- A system may have **no solution**, may have **unique solution** (nghiệm duy nhất), or may have an **infinite family of solutions** (vô số nghiệm)

- A system that has **no solution** is called **inconsistent** (không tương thích)
- A system that has **at least one solution** is called **consistent** (tương thích)

Inconsistent (không tương thích)	**Consistent (tương thích)**	
No solutions (vô nghiệm)	**Unique solution (nghiệm duy nhất)**	**Infinitely many solutions (vô số nghiệm)**

Example 1

- $\begin{cases} x+2y=1 \\ x+2y=3 \end{cases}$ no solution **Inconsistent**

- $\begin{cases} x+y-z=1 \\ x+y+z=3 \end{cases}$ (0,2,1), (2,0,1) (t,2-t,1) **Consistent** (infinitely many solutions)

(t,2-t,1) is called a **general solution** and given in **parametric form** , *t* is **parameter** **(*t* is arbitrary)**

Algebraic Method

$$\begin{aligned} 3x_1 + 2x_2 - x_3 + x_4 &= -1 \\ 2x_1 - x_3 + 2x_4 &= 0 \\ 3x_1 + x_2 + 2x_3 + 5x_4 &= 2 \end{aligned}$$

$$\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

constant matrix
(ma trận cột tự do)

$$\left\langle \begin{array}{cccc|c} 3 & 2 & -1 & 1 & -1 \\ 2 & 0 & -1 & 2 & 0 \\ 3 & 1 & 2 & 5 & 2 \end{array} \right\rangle$$

augmented matrix
(ma trận mở rộng)

$$\left\langle \begin{array}{cccc} 3 & 2 & -1 & 1 \\ 2 & 0 & -1 & 2 \\ 3 & 1 & 2 & 5 \end{array} \right\rangle$$

coefficients matrix
(ma trận hệ số)

Elementary Operations
(biến đổi sơ cấp)

Difficult $\xrightarrow{\text{Equivalent}}$ **Easy**

Two systems are said to be **equivalent** if the have the same set of solutions

Elementary Operations
(phép biến đổi sơ cấp)

- **Interchange** two equations (I)
- **Multiply** one equation by a **nonzero number** (II)
- **Add a multiple** of one equation to a different equation (III)

Type I (interchange two rows)

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & 0 \\ 4 & 1 & 6 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 4 & 1 & 6 \\ 2 & 5 & 0 \\ 1 & 2 & -3 \end{bmatrix}$$

Type II (multiply one row by a nonzero number)

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & 0 \\ 4 & 1 & 6 \end{bmatrix} \xrightarrow{r_1 \to 3r_1} \begin{bmatrix} 3 & 6 & -9 \\ 2 & 5 & 0 \\ 4 & 1 & 6 \end{bmatrix}$$

Type III (add a multiple of one row to another row)

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & 0 \\ 4 & 1 & 6 \end{bmatrix} \begin{matrix} r_2 \to r_2 - 2r_1 \\ r_3 \to r_3 - 4r_1 \\ \longrightarrow \end{matrix} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 6 \\ 0 & -7 & 18 \end{bmatrix}$$

Example 2

Consider the system $\begin{cases} x - y = 1 \\ x + 2y = -2 \end{cases}$

augmented matrix $\left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 1 & 2 & -2 \end{array} \right\rangle$

Example 2

$$\begin{cases} x-y=1 \\ x+2y=-2 \end{cases} \Leftrightarrow \begin{cases} x-y=1 \\ 0x+3y=-3 \end{cases} \Leftrightarrow \begin{cases} x-y=1 \\ 0x+y=-1 \end{cases} \Leftrightarrow \begin{cases} x+0y=0 \\ 0x+y=-1 \end{cases}$$

$$\left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 1 & 2 & -2 \end{array} \right\rangle \overset{(III)}{\rightarrow} \left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & -3 \end{array} \right\rangle \overset{(II)}{\rightarrow} \left\langle \begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & -1 \end{array} \right\rangle \overset{(III)}{\rightarrow} \left\langle \begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1 \end{array} \right\rangle$$

$\Rightarrow$Solution (0,-1)

Theorem. *Suppose an elementary operation is performed on a system of linear equations. Then the resulting system has the* ***same set of solutions*** *as the original system, so the two system are equivalent (tương đương)*

In this case,their augmented matrices are called ***row-equivalent***

Elementary (row) Operations on matrix

In case of two equations in two variables, the goal is to produce a matrix of the form

$$\left\langle \begin{array}{cc|c} 1 & 0 & * \\ 0 & 1 & * \end{array} \right\rangle$$

In case of three equations in three variables, the goal matrix is

$$\left\langle \begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array} \right\rangle$$

Example 4: Find all solutions to the following system of equations

$$3x + 4y + z = 1$$
$$2x + 3y = 0$$
$$4x + 3y - z = -2$$

Solution

The augmented matrix of the original system is

$$\left\langle \begin{array}{ccc|c} 3 & 4 & 1 & 1 \\ 2 & 3 & 0 & 0 \\ 4 & 3 & -1 & -2 \end{array} \right\rangle$$

Subtracting row 2 from row 1, the result is

$$\left\langle \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 3 & 0 & 0 \\ 4 & 3 & -1 & -2 \end{array} \right\rangle$$

Subtract 2 times row 1 from row 2 and subtract 4 times row 1 from row 3, the result is

$$\left\langle \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -2 \\ 0 & -1 & -5 & -6 \end{array} \right\rangle$$

Subtracting row 2 from 1 and adding row 2 to row 3, the result is

$$\left\langle \begin{array}{ccc|c} 1 & 0 & 3 & 3 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & -7 & -8 \end{array} \right\rangle$$

Begin by multiplying row 3 by $-\frac{1}{7}$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 3 \\ 0 & 1 & -2 & -2 \\ 0 & 0 & 1 & \frac{8}{7} \end{array}\right]$$

Now subtract 3 times row 3 from row 1, and add 2 times row 3 to row 2 to get

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{3}{7} \\ 0 & 1 & 0 & \frac{2}{7} \\ 0 & 0 & 1 & \frac{8}{7} \end{array}\right]$$

The corresponding equations are $x = -\frac{3}{7}$, $y = \frac{2}{7}$, and $z = \frac{8}{7}$, which give the (unique) solution.

1.2. Gaussian Elimination (phép khử Gauss)

By using elementary row operations to carry the augmented matrix to "nice" matrix such as

$$\left\langle \begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array} \right\rangle$$

(and it is said to be in **row- echelon form (dạng bậc thang theo dòng))**

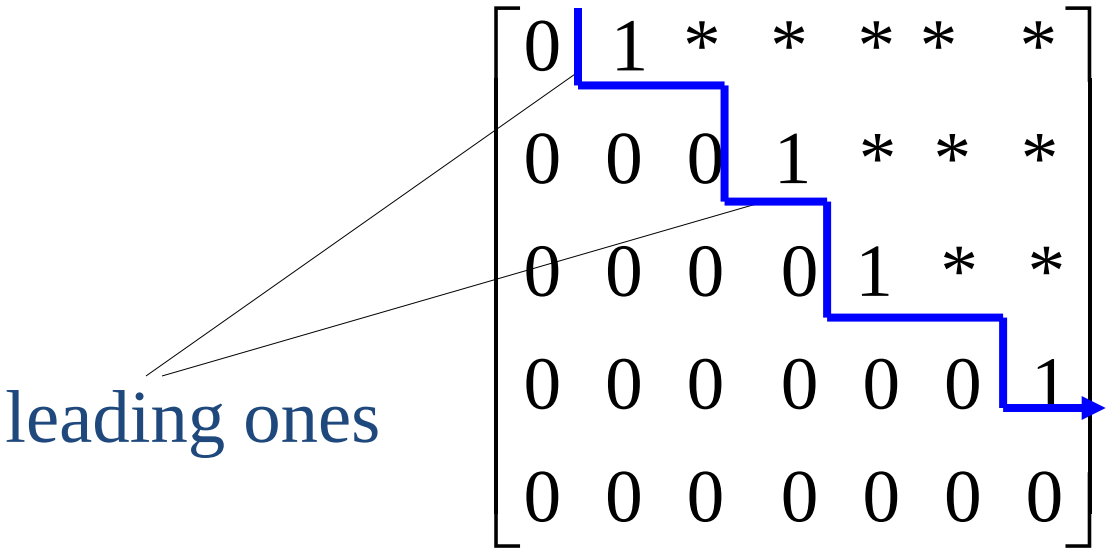
A row-echelon matrix has 3 properties

$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- All the zero rows are at the bottom
- The first nonzero entry from the left in each nonzero row is a 1, called the **leading 1** for that row
- Each leading 1 is **to the right** of all leading 1's in the rows above it

Row-echelon matrix

The row-echelon matrix has the "staircase" form



$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(for any choice in *-position)

Row-echelon matrix

$$\begin{bmatrix} 1 & * & * \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$$

~~$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 1 & * & * \end{bmatrix}$$~~

$$\begin{bmatrix} 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

~~$$\begin{bmatrix} 1 & * & * \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$~~

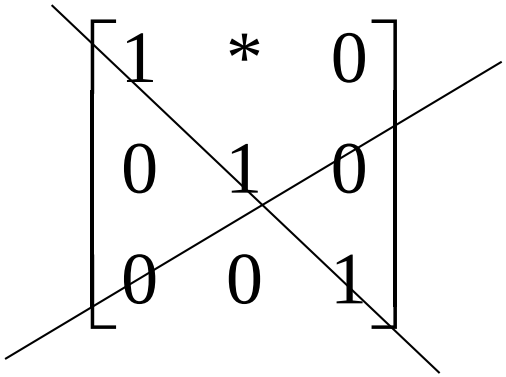
A reduced row-echelon matrix
has the properties

- It is a row-echelon matrix
- Each leading 1 is the **only nonzero** entry in its column

$$\begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

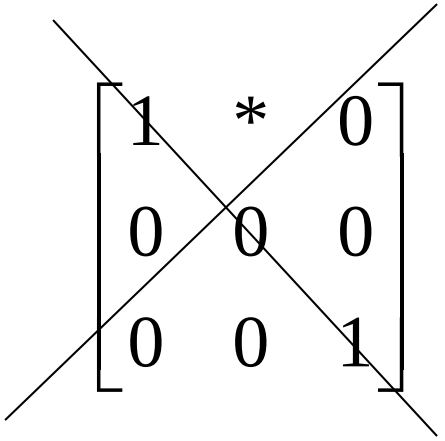
Reduced row- echelon matrix

$$\begin{bmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Find x,y so that the matrix

$$\begin{bmatrix} 1 & x & y \\ 0 & x & 1 \end{bmatrix}$$

- is row-echelon matrix
- is **reduced row-echelon** matrix

Gaussian Algorithm

Theorem. *Every matrix can be brought to (reduced) row-echelon form by a series of elementary row operations*

Step 1. If all row are **zeros**, stop

Step 2. Otherwise, find the first column from the left containing a nonzero entry (call it *a*) and move the row containing *a* to the top position

Step 3. Multiply that row by 1/a to create the **leading 1**

Step 4. By subtracting multiples of that row from the rows below it, make each entry below the leading 1 **zero**

Step 5. Repeat step 1-4 on the matrix consisting of the remaining rows

Gaussian Algorithm

$$step\ 1: \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow stop$$

leading one

$$\begin{bmatrix} 0 & 0 & 1 & 3 \\ 2 & 2 & -1 & 0 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 2 & 2 & -1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{r_1 \rightarrow \frac{1}{2} r_1} \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 4 & 7 & 5 & -1 \end{bmatrix} \xrightarrow{r_4 \rightarrow r_4 - 4r_1} \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 3 & 0 & 6 \\ 0 & 3 & 7 & -1 \end{bmatrix}$$

step 2 step 3 step 4 step 5

Example

Carry the matrix

$$\begin{bmatrix} 2 & 6 & -2 & 2 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix}$$

to row-echelon matrix

to reduced row-echelon matrix

$$\begin{bmatrix} 2 & 6 & -2 & 2 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix} \xrightarrow{r_1 \to \frac{1}{2} r_1} \begin{bmatrix} 1 & 3 & -1 & 1 \\ -2 & -3 & 11 & 4 \\ 3 & 11 & 3 & 0 \end{bmatrix} \xrightarrow[r_3 \to r_3 - 3r_1]{r_2 \to r_2 + 2r_1} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 3 & 9 & 6 \\ 0 & 2 & 6 & -3 \end{bmatrix}$$

$$\xrightarrow{r_2 \to \frac{1}{3} r_2} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 2 & 6 & -3 \end{bmatrix} \xrightarrow{r_3 \to r_3 - 2r_2} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & -7 \end{bmatrix} \xrightarrow{r_3 \to -\frac{1}{7} r_3} \begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

row-echelon matrix

$$\begin{bmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} r_2 \to r_2 - 2r_3 \\ r_1 \to r_1 - r_3 \\ \to \end{matrix} \begin{bmatrix} 1 & 3 & -1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{matrix} r_1 \to r_1 - 3r_2 \\ \to \end{matrix} \begin{bmatrix} 1 & 0 & -10 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

reduced row-echelon matrix

Gauss-Jordan Elimination

(for solving a system of linear equations)

Step 1. Using elementary row operations, **augmented matrix→reduced row-echelon matrix**

$$\begin{bmatrix} 1 & 0 & -10 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{cases} x + 0y - 10z = 0 \\ 0x + 1y + 3z = 0 \\ 0x + 0y + 0z = 1 \end{cases}$$

reduced row echelon matrix

Step 2. If a row [0 0 0…0 1] occurs, the system is **inconsistent**

Step 3. Otherwise, assign the **nonleading variables** as parameters, solve for the leading variables in terms of parameters

$$\begin{bmatrix} 1 & 0 & -10 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} 1x + 0y - 10z = 0 \\ 0x + 1y + 3z = 0 \\ 0x + 0y + 0z = 0 \end{cases}$$

z=t (parameter) **z is nonleading variable**

$$\begin{bmatrix} 1 & 0 & -10 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{cases} 1x + 0y - 10z = 0 \\ 0x + 1y + 3z = 0 \\ 0x + 0y + 0z = 1 \end{cases}$$

reduced row echelon matrix

Example 2

Solve the following system of equations.

$$x + 10z = 5$$
$$3x + y - 4z = -1$$
$$4x + y + 6z = 1$$

$$\begin{bmatrix} 1 & 0 & 10 & 5 \\ 3 & 1 & -4 & -1 \\ 4 & 1 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 1 & -34 & -19 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 10 & 5 \\ 0 & 1 & -34 & -16 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

row echelon matrix

inconsistent

Example 3: Solve the following system of equations.

$$x_1 - 2x_2 - x_3 + 3x_4 = 1$$
$$2x_1 - 4x_2 + x_3 = 5$$
$$x_1 - 2x_2 + 2x_3 - 3x_4 = 4$$

leading one

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

row echelon matrix

x_2, x_4 are nonleading variables, so we set $x_2 = t$ and $x_4 = s$ (parameters) and then compute x_1, x_3

Example

Which condition on the numbers a,b,c is the system

$$\begin{cases} 3x + y - z = a \\ x - y + 2z = b \\ 5x + 3y - 4z = c \end{cases}$$

consistent ? unique solution?

The rank of a matrix

- The reduced row-echelon form of a matrix A is uniquely determined by A, but the row-echelon form of A is not unique
- The number r of leading 1's is the same in each of the different row-echelon matrices
- As r depends only on A and not on the row-echelon forms, it is called **the rank of the matrix A**, and written r=rankA

The rank of a matrix

- If the a matrix A has the row-echelon matrix is

$$\begin{bmatrix} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

then rankA=4

Example 5

Compute the rank of $A = \begin{bmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & -5 & 8 \end{bmatrix}$.

Solution

The reduction of A to row – echelon form is

$$A = \begin{bmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & -5 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 4 \\ 0 & -1 & 5 & -8 \\ 0 & 1 & -5 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 4 \\ 0 & 1 & -5 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Because this row – echelon matrix has two leading 1's, rank A=2.

Theorem 2

Suppose a system of m equations in **n variables** has a solution. If the **rank** of the augment matrix is **r** then the set of solutions involves exactly **n-r** parameters

leading one

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

rankA=2

4(number of variables)- 2(rankA) =2
(two parameters : $x_2=t$, $x_4=s$)

1.3.Homogeneous Equations (phương trình thuần nhất)

- The system is called **homogeneous** (thuần nhất) if the constant matrix has all the entry are zeros

- Note that every homogeneous system **has at least one solution (0,0,…,0),** called **trivial solution** (nghiệm tầm thường)

- If a homogeneous system of linear equations has **nontrivial solution** (nghiệm không tầm thường) then it has infinite family of solutions (vô số nghiệm)

Example 1: Show that the following homogeneous system has nontrivial solutions.

$$x_1 - x_2 + 2x_3 + x_4 = 0$$
$$2x_1 + 2x_2 - x_4 = 0$$
$$3x_1 + x_2 + 2x_3 + x_4 = 0$$

Solution

The reduction of the augmented matrix to reduced row – echelon form is outined below.

$$\left\langle\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 2 & 2 & 0 & -1 & 0 \\ 3 & 1 & 2 & 1 & 0 \end{array}\right\rangle \rightarrow \left\langle\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 4 & -4 & -3 & 0 \\ 0 & 4 & -4 & -2 & 0 \end{array}\right\rangle \rightarrow \left\langle\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right\rangle$$

The leading variables are x_1, x_2 and x_4, so x_3 is assigned as a parameter – say $x_3 = t$. Then the general solution is

$$x_1 = -t$$
$$x_2 = t$$
$$x_3 = t$$
$$x_4 = 0$$

Hence, taking $t = 1$, we get a nontrivial solution: $x_1 = -1, x_2 = 1, x_3 = 1, x_4 = 0$.

Theorem 1

If a homogeneous system of linear equations has **more variables than equations**, then it has nontrivial solution (in fact, infinitely many)

Note that the converse of theorem 1 is not true

Summary

System of	Inconsistent (no solutions)	Consistent	
		Unique solution (exactly one solution)	Infinitely many solutions
linear equations	yes	yes	yes
linear equations that has more variables than equations	Yes	no	yes
homogeneous linear equations	No	yes	yes
homogeneous linear equations that has more variables than equations	No	no	yes

P16: 1- 7

P24: 2,5

7 and 9c: P17